

S30 — Probability: Conditional, Bayes & Business Analysis

1 | Conditional Probability — Customer Purchase

Customer Source	Customers	Purchased
Advertisement	1,000	200
Organic	1,500	300

1. Total customers.
2. Total purchasers.
3. $P(\text{Purchased})$
4. $P(\text{Purchased} \mid \text{Advertisement})$
5. $P(\text{Advertisement} \mid \text{Purchased})$
6. Q4 aur Q5 mein denominator kya hai?
7. Q4 aur Q5 ka meaning simple English mein explain karo.
8. Kaunsi probability higher hai?

2 | Independent vs Dependent — Business

A company observes whether customers:

- Saw an advertisement
 - Purchased a product
1. If seeing an advertisement changes the probability of purchasing, are the events independent or dependent?
 2. If purchase probability is exactly the same whether or not the customer saw the advertisement, are they independent or dependent?
 3. Explain why correlation/association should be investigated before claiming that advertisement causes purchase.

4. Give one example of two independent events.
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3 | Addition Rule — Customer Channels

A company has 200 customers:

- 120 use the Mobile App.
- 80 use the Website.
- 30 use both.

1. $P(App)$
 2. $P(Website)$
 3. $P(App \cap Website)$
 4. $P(App \cup Website)$
 5. Number of customers using at least one channel.
 6. Number of customers using neither channel.
 7. Why do we subtract the "Both" customers?
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4 | Complement Rule — Churn

A subscription company's churn probability is **12%**.

1. Find the probability of **not churning**.
 2. Convert it to percentage.
 3. If there are 1,000 customers, approximately how many might not churn?
 4. Does this guarantee that exactly that many customers will stay?
 5. Explain **probability vs guarantee**.
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5 | Mutually Exclusive vs Overlapping

A six-sided die is rolled.

- A = getting an even number.
- B = getting an odd number.
- C = getting a number greater than 3.

1. Write A.
2. Write B.
3. Write C.
4. Is A and B mutually exclusive?
5. Find $A \cap B$.
6. Is A and C mutually exclusive?
7. Find $A \cap C$.
8. Explain why A and C overlap.

6 | Bayes — Fraud Detection

Transaction Type	Transactions	Fraud
Online	8,000	240
Offline	2,000	40

1. Total transactions.
2. Total fraud transactions.
3. $P(\text{Fraud})$
4. $P(\text{Fraud} \mid \text{Online})$
5. $P(\text{Online} \mid \text{Fraud})$
6. Which is higher: Q4 or Q5?
7. What does the denominator of Q5 represent?

8. If a transaction is known to be fraudulent, what probability are we finding in Q5?
 9. Does $P(\text{Online} \mid \text{Fraud})$ mean that every online transaction is fraudulent?
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7 | Rate vs Count — Marketing

Campaign	Customers Reached	Purchased
A	8,000	640
B	2,000	300

1. Campaign A conversion rate.
 2. Campaign B conversion rate.
 3. Which campaign has the higher conversion rate?
 4. Which campaign has more purchasers?
 5. Are "higher conversion rate" and "more purchasers" the same thing?
 6. If management wants the **highest efficiency**, which campaign should they investigate?
 7. If management wants the **largest number of purchasers**, which campaign should they investigate?
 8. Explain why customer count matters.
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8 | Find the Wrong Analyst Interpretation

An analyst says:

"Campaign A has 640 purchasers and Campaign B has 300 purchasers. Therefore, Campaign A is definitely better."

1. Is this conclusion correct?
2. What additional metric should be compared?
3. Calculate the conversion rate of both campaigns.
4. Why can a campaign have more purchasers but a lower conversion rate?

5. Rewrite the analyst's conclusion correctly.

9 | Reverse Probability — Employee Skills

Department	Employees	Python Skilled
Data Analytics	250	150
Finance	500	100
Marketing	750	75

1. Total employees.
 2. Total Python-skilled employees.
 3. $P(\text{Python Skilled})$
 4. $P(\text{Python Skilled} \mid \text{Data Analytics})$
 5. $P(\text{Data Analytics} \mid \text{Python Skilled})$
 6. Which department has the highest Python-skilled **rate**?
 7. Which department has the largest number of Python-skilled employees?
 8. Are Q6 and Q7 asking the same question?
 9. Explain the difference between **rate** and **count**.
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10 | Full Data Analyst Case — Subscription Business

Plan	Customers	Churned
Basic	4,000	600
Premium	2,000	160
Enterprise	1,000	50

1. Total customers.
2. Total churned customers.
3. Overall churn probability.
4. $P(\text{Churned} \mid \text{Basic})$

5. $P(\text{Churned} \mid \text{Premium})$
6. $P(\text{Churned} \mid \text{Enterprise})$
7. $P(\text{Basic} \mid \text{Churned})$
8. Which plan has the highest **churn rate**?
9. Which plan has the largest **number of churned customers**?
10. Are Q8 and Q9 asking the same question?
11. Which plan should management investigate if the goal is to reduce the **highest churn rate**?
12. Which plan should management investigate if the goal is to reduce the **largest number of churned customers**?
13. Can we conclude that Basic is **causing** churn?
14. Give **two additional pieces of information** you would investigate.
15. Explain why looking only at the number of churned customers can be misleading.
16. Explain why looking only at churn rate can also be misleading.